No.	Status	Date
HIS-2010-0318-DS-3-RV-101	Issued	2010-09-13

UT Thickness Data Sheet - 3-RV-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 3-RV-101, located within the ATU-100. The inspection was conducted as part of the 3-year regenerator overhead survey detailed in HIS-2010-0318.

Kevin Doucette, acting as lead UT technician (SNT-TC-1A UT Level II) for Halward Inspection Services Ltd., performed the measurements and is responsible for the accuracy of the recorded data.

Technical Specifications

The subject line, 3-RV-101, is constructed from A106 Gr.B with an outer diameter of 88.9 mm and a nominal wall thickness of 7.62 mm. The design limits for this circuit are 345 kPa(g) and 121 °C.

The minimum allowable thickness (t-min) is 3.56 mm. Any reading recorded below this retirement thickness shall be raised as an NCR.

Procedural Compliance

Thickness measurements were performed in accordance with QSCP-NDT-001 Rev 1, utilizing an ATU-100 gauge with waveform display. This procedure requires CMLs at fittings and scanning at elbows and tees.

The work was also performed under HIS-UT-002 Rev 1 and conforms to the requirements of ASTM E797.

Asset Integrity Analysis

Corrosion rates are determined by the larger of the long-term rate or the short-term rate over the latest pair of readings at least 3 years apart. Remaining life is calculated as (t - t-min) / rate.

As a Class 1 circuit (sour gas, regenerator overhead), the maximum interval for thickness measurements is 5 years, though the regenerator overhead circuit is currently on a 3-year thickness interval.

Thickness readings, measured 2010-09-13

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2007-09-18	6.95	6.78	0.056	0.044	>50	2013-09-13	
3-RV-101-02	7.62	3.56	2007-09-18	6.89	6.59	0.068	0.075	40.6	2013-09-13	
3-RV-101-03	7.62	3.56	2007-09-18	6.56	6.16	0.094	0.090	27.8	2013-09-13	
3-RV-101-04	7.62	3.56	2007-09-18	6.66	6.43	0.080	0.048	35.9	2013-09-13	
3-RV-101-05	7.62	3.56	2007-09-18	6.87	6.63	0.072	0.084	36.8	2013-09-13	
3-RV-101-06	7.62	3.56	2007-09-18	6.71	6.43	0.080	0.068	35.9	2013-09-13	
3-RV-101-07	7.62	3.56	—	—	6.20	—	—	—	2013-09-13	
3-RV-101-08	7.62	3.56	—	—	6.71	—	—	—	2013-09-13	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2007-09-18	4.83	4.55	0.068	0.066	26.1	2013-09-13	
2-RV-102-02	5.54	2.79	2007-09-18	4.67	4.60	0.068	0.053	26.5	2013-09-13	
2-RV-102-03	5.54	2.79	2007-09-18	5.01	4.84	0.044	0.066	31.1	2013-09-13	
2-RV-102-04	5.54	2.79	2007-09-18	5.12	4.84	0.051	0.051	40.2	2013-09-13	
2-RV-102-05	5.54	2.79	—	—	5.04	—	—	—	2013-09-13	
2-RV-102-06	5.54	2.79	—	—	4.82	—	—	—	2013-09-13	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-03	8.56	3.81	—	—	8.59	—	—	—	2020-09-13	
4-RA-104-04	8.56	3.81	—	—	8.52	—	—	—	2020-09-13	
4-RA-104-05	8.56	3.81	—	—	8.43	—	—	—	2020-09-13	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2010-09-13	_____